Introductory Econometrics with Recitation — Quiz 4

ECON2023

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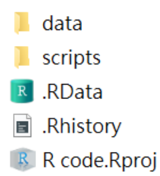
( ) 1. (**2 points**) Alice creates an R Markdown file named `wk_8.Rmd` in her project folder. Suppose Alice opens the RStudio project `R_code` and then opens the R Markdown file `wk_8.Rmd`. She wants to import the file `county.csv` from the `data` folder in an R Markdown code chunk. Which of the following lines of R code will do the job?



A. `county <- read.csv("county.csv")`

B. `county <- read.csv("data/county.csv")`

C. `county <- read.csv("../data/county.csv")`

D. `county <- read.csv("~/data/county.csv")`

( ) 2. (**2 points**) Based on the previous question, suppose Alice opens a new R script `wk_7.R` in the `script` folder of the project and wants to source the R script `calc_win.R`, which is in the same `script` folder. Which of the following lines of R code will do the job?

A. `source(here("calc_win.R"))`

B. `source(here("script", "calc_win.R"))`

C. `source(here("../", "calc_win.R"))`

D. `source(here("../script", "calc_win.R"))`

( ) 3. (**2 points**) Which of the following statements is FALSE?

A. Applying `na.omit()` will remove any row in which at least one column contains `NA`.

B. The following code extracts the numeric parts `"12"`, `"34"`, and `"56"`.

```
substring(c("AB12", "CD34", "EF56"), 3, 4)
```

C. We can use `colnames()` to assign column names to a matrix.

D. The output of `class(ls())` is `"list"`.

( ) 4. (**2 points**) What is the output of the following code?

```
nchar(c(-15.6, 10, FALSE))
```

A.

```
5 2 0
```

B.

```
5 2 1
```

C.

```
5 2 3
```

D.

```
5 2 5
```

( ) 5. (**2 points**) Which of the following statements is TRUE?

A. We can use `table(..., addmargins = ...)` to add a row of column sums and a column of row sums to the table.

B. If more than two vectors are passed to `cor()`, the output will be saved as a data frame.

C. The output of `dim(df)` is the same as `c(ncol(df), nrow(df))`.

D. There exists a numeric vector `vec` such that the following two pieces of code generate different outputs:

```
vec %>% scale(center = FALSE, scale = TRUE) %>%
        scale(center = TRUE, scale = FALSE)
```

```
vec %>% scale(center = TRUE, scale = FALSE) %>%
        scale(center = FALSE, scale = TRUE)
```

( ) 6. (**2 points**) Which of the following statements about `lm()` and `lm_robust()` is TRUE?

A. The objects returned by `lm()`, `lm_robust()`, and `summary(lm(...))` are lists.

B. The standard errors calculated by `lm_robust()` are always smaller than those calculated by `lm()`, so `lm_robust()` gives wider and more conservative confidence intervals.

C. The printed output layouts from `lm()` and `lm_robust()` are the same; only the numbers differ, while `summary()` shows more detailed results.

D. `lm()` fails to obtain correct standard errors when homoskedasticity holds.

( ) 7. (**2 points**) Below is a regression table from `screenreg()`. Which of the following statements about the table is TRUE?

```
=============================================================
              Single        Multiple      Multiple (Robust SE)
-------------------------------------------------------------
(Intercept)   -6.095 ***   -6.043 ***   -6.043 ***
              (0.465)      (0.480)      (0.529)
weeks          0.344 ***    0.341 ***    0.341 ***
              (0.012)      (0.013)      (0.014)
visits                      0.007        0.007
                           (0.009)      (0.009)
-------------------------------------------------------------
R^2            0.449        0.434        0.434
Adj. R^2       0.448        0.433        0.433
Num. obs.        998          990          990
RMSE                                     1.121
=============================================================
*** p < 0.001; ** p < 0.01; * p < 0.05
```

A. `***` next to a coefficient means that the p-value for that coefficient is between 0.01 and 0.001.

B. The table is printed with the argument `include.ci = TRUE`.

C. To print such a table, we can directly use the objects returned by `lm()` and `lm_robust()` without first applying `summary()`.

D. From the table, we can infer that `weeks` is uncorrelated with `visits`.

(   ) 8. (**2 points**) Which of the following numbers is most likely closest to the output below?

```
vec <- c()
for(i in 1:10000){
  vec <- c(vec, mean(runif(n = 10, min = c(20, 80), max = c(100, 200, 300))))
}
mean(vec)
```

A. 100

B. 105

C. 120

D. 125

(   ) 9. (**2 points**) Which of the following statements about random number generators (RNG) is FALSE?

A. For `rbinom()`, there are two arguments that affect its distribution: `size` and `prob`.

B. The expected value of a random number generated by `rexp(n = 1, rate = 5)` is $\frac{1}{5}$.

C. `set.seed()` sets R's random-number generator state. We set a seed to ensure that the code produces the same sequence every time it is rerun.

D. Without setting a seed, the seed depends on the current time. Therefore, if two students run the same RNG program on different laptops at exactly the same time, they will generate the same output.

( ) 10. (**2 points**) Below is the setup chunk you will see after opening a new R Markdown file. Which of the following statements about it is FALSE?

````
```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```
````

A. `include = FALSE` means that this code will not be run or shown after knitting. To make it work, we have to set it to `TRUE`.

B. The symbol `::` allows us to access an object from a package without using `library()`.

C. `opts_chunk` is a list in the `knitr` package.

D. `setup` is the chunk label; deleting it will not affect the functioning of the chunk.

( ) 11. (**2 points**) Consider the following two ways to generate `kaiji_stress`:

```
# Code 1
side <- ifelse(runif(50) < 0.5, "emperor_side", "slave_side")
kaiji_stress <- rnorm(
  n = 50,
  mean = ifelse(side == "slave_side", 0.5, 0.7),
  sd = 0.4
)
```

```
# Code 2
side_rand <- runif(50)
side <- ifelse(side_rand < 0.5, "emperor_side", "slave_side")

kaiji_stress <- ifelse(
  side == "slave_side",
  rnorm(50, 0.5, 0.4),
  rnorm(50, 0.7, 0.4)
)
```

Which of the following statements is FALSE?

A. In both pieces of code, each observation has a 50% probability of being assigned to `emperor_side` and a 50% probability of being assigned to `slave_side`.

B. In Code 1, the code would fail if the length of `side` did not match `n = 50`. Therefore, a cleaner way is to define `round <- 50` and replace `50` with `round` to avoid such an error.

C. In Code 2, R generates 50 random draws with `mean = 0.5` and 50 random draws with `mean = 0.7`. Then, for each observation, it selects which draw to use for `kaiji_stress[i]` according to the value of `side` in the `i`-th round.

D. Even when the two pieces of code aim to perform similar jobs and use the same random seed, they can still produce different simulated values.

Below is the context and part of the answer code for Week 6's practices. Please read the content and answer questions 12-15:

Context: Lawrence and Holo goes from town to town to make profits as a travelling merchant. To decide where bread is relatively expensive or cheap, Lawrence records the quoted market prices of several goods in 24 towns and cities along the route. The dataset contains:

- `place`: The name of the town or city.
- `region`: The broad trading region of the market (`North`, `Central`, or `South`).
- `market_type`: The type of market (`village`, `market_town`, `church_city`, `port_city`, or `spa_town`).
- `bread_price`: The quoted bread price in each market.
- `wheat_price`: The quoted wheat price in each market.
- `rye_price`: The quoted rye price in each market.
- `cheese_price`: The quoted cheese price in each market.
- `butter_price`: The quoted butter price in each market.

Prices are recorded as character strings, and some markets use the currency unit `trenni` while others use `lumione`. The exchange rate is 1 `lumione` = 4 `trenni`. **In the data, the price are saved as the string with the form "{number}{currency}".** For example, `11.1trenni` or `2.3lumione`.

In this world, rye is often treated as a substitute for wheat, while cheese and butter are commonly sold and consumed together with bread. Hence, Lawrence wants to know how the price of `bread` is related to the prices of `wheat`, `rye`, `cheese`, and `butter`. Follow the instructions below to help Lawrence study these price relationships.

Answer code 1:

```
for(i in 1:nrow(spice_and_wolf_tidy)){
  if(grepl("trenni", spice_and_wolf_tidy[i, 4])){
    spice_and_wolf_tidy[i, 4:8] <- substring(
      spice_and_wolf_tidy[i, 4:8],
      1,
      nchar(spice_and_wolf_tidy[i, 4:8]) - 6
      ) %>% as.numeric()
  }
  else{
    spice_and_wolf_tidy[i, 4:8] <- (substring(
```

```
      spice_and_wolf_tidy[i, 4:8],
      1,
      nchar(spice_and_wolf_tidy[i, 4:8]) - 7
      ) %>% as.numeric()
      ) * 4
  }
}
```

Answer code 2:

```
for (i in 4:8) {
  spice_and_wolf_tidy[[i]] <- ifelse(
    grepl("trenni", spice_and_wolf_tidy[[i]]),
    as.numeric(substring(spice_and_wolf_tidy[[i]], 1,
                         nchar(spice_and_wolf_tidy[[i]]) - 6)),
    as.numeric(substring(spice_and_wolf_tidy[[i]], 1,
                         nchar(spice_and_wolf_tidy[[i]]) - 7)) * 4
  )
}
```

( ) 12. (**2 points**) In Answer Code 1, the syntax `spice_and_wolf_tidy[i, 4:8]` is used. What is the primary reason this code might be considered "risky"?

A. The code will only work if the dataset has exactly 8 columns.

B. It is impossible to perform mathematical operations on a range of columns at once in R.

C. If Lawrence decides to add a new column before the `bread_price` column, the code will target the wrong variables.

D. Using numeric indices forces R to convert the entire data frame into a matrix, which slows down the calculation.

( ) 13. (**2 points**) After running each code snippet separately, Lawrence uses the `class()` function to check the data type of the `bread_price` column. Which of the following correctly describes the resulting data types for Answer Code 1 and Answer Code 2, respectively?

A. `numeric`, `numeric`

B. `character`, `numeric`

C. `numeric`, `character`

D. `character`, `character`

( ) 14. (**2 points**) Suppose some markets use both `trenni` and `lumione` for different goods. Which of the following statements about the two pieces of code is TRUE?

A. Answer Code 1 is more efficient because the `if` statement ensures that the currency is checked only once per row for all goods.

B. Answer Code 2 is better because Answer Code 1 assumes that all goods in a row share the same currency based on a single check, while Answer Code 2 evaluates the currency of each cell independently.

C. Answer Code 2 will fail because `ifelse()` requires all prices in a column to have the same currency in order to perform `substring()` correctly.

D. Both pieces of code will correctly handle mixed currencies within a row because `grepl()` and `substring()` are naturally vectorized across multiple columns.

( ) 15. (**2 points**) A student tries to clean the data using the following code:

```
for (i in 4:8) {
  if (grepl("trenni", spice_and_wolf_tidy[[i]])) {
    spice_and_wolf_tidy[[i]] <- as.numeric(
      substring(spice_and_wolf_tidy[[i]],
                1,
                nchar(spice_and_wolf_tidy[[i]]) - 6))
  } else {
    spice_and_wolf_tidy[[i]] <- as.numeric(
      substring(spice_and_wolf_tidy[[i]],
                1,
                nchar(spice_and_wolf_tidy[[i]]) - 7)) * 4
  }
}
```

Upon running this code, R throws the following error:

```
Error in if (grepl("trenni", spice_and_wolf_tidy[[i]])) { :
  the condition has length > 1
```

Which of the following best explains why this error occurred?

A. The `if` statement is designed to evaluate a single logical value, but `grepl()` returns a logical vector.

B. The code checks a whole price column at once, so R automatically chooses the first row to decide whether the whole column is in `trenni` or `lumione`.

C. The error occurs because some rows contain `trenni` and some rows contain `lumione`. If all rows used the same currency, the `if` statement would work correctly.

D. The error occurs because `grepl()` returns the positions of the rows containing `trenni`, but `if` can only use `TRUE`/`FALSE` values.